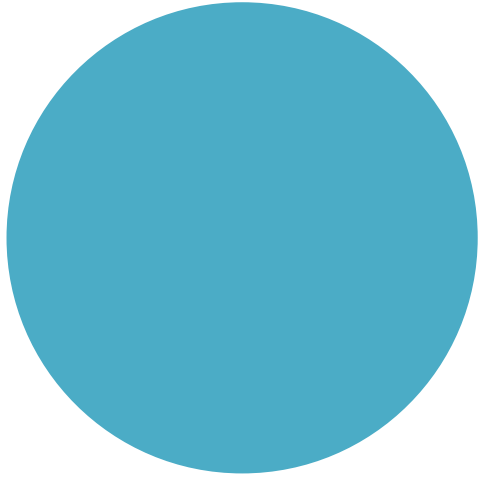


Mobile Devices and Applications

Unit 12 – Location
Providers & Maps

Summary



Fetch user location



Reverse geocoding using Geocoder



MapFragment



Showing current location



Displaying markers



Map controls

**FETCH USER CURRENT
LOCATION**

Google API Client

```
FusedLocationProviderClient fusedLocationClient = LocationServices.getFusedLocationProviderClient( activity: this);  
//1. getLastLocation  
fusedLocationClient.getLastLocation()  
    .addOnSuccessListener( activity: this, new OnSuccessListener<Location>() {  
        @Override  
        public void onSuccess(Location location) {  
            // Got last known location. In some rare situations this can be null.  
            if (location != null) {  
                // Logic to handle location object  
                LatLng newLoc = new LatLng(location.getLatitude(), location.getLongitude());  
                mMap.addMarker(new MarkerOptions().position(newLoc).title("New Update"));  
                mMap.moveCamera(CameraUpdateFactory.newLatLng(newLoc));  
            }  
        }  
    });
```

USER LOCATION

- Location service: Context.LOCATION_SERVICE
 - Used for creating a **LocationManager** object
- Location is determined using different sources such as:
 - GPS receiver;
 - WiFi networks;
 - Mobile cell phone;
- Permissions
 - android.permission.ACCESS_FINE_LOCATION
 - android.permission.ACCESS_COARSE_LOCATION

USER LOCATION

- Google Play Service evaluation and Location Service initialization;
- Initialization of a *listener* by specifying:
 - The source used for getting the location;
 - The amount of time between two different location queries;
- Notifications received when:
 - location changes;
 - the source providing data changes state;

USER LOCATION

Add marker

```
@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;

    // Add a marker in Sydney and move the camera
    LatLng sydney = new LatLng(-34, 151);
    mMap.addMarker(new MarkerOptions().position(sydney).title("Marker in Sydney"));
    mMap.moveCamera(CameraUpdateFactory.newLatLng(sydney));
}
```

Current
location

```
protected void createLocationRequest() {
    LocationRequest mLocationRequest = new LocationRequest();
    mLocationRequest.setInterval(10000);
    mLocationRequest.setFastestInterval(5000);
    mLocationRequest.setPriority(LocationRequest.PRIORITY_HIGH_ACCURACY);
}
```

Location Manager / Provider

```
LocationManager locationManager =  
    (LocationManager) this.getSystemService(Context.LOCATION_SERVICE);
```

```
LocationProvider provider =  
    locationManager.getProvider(LocationManager.GPS_PROVIDER);
```

f	EXTRA_PROVIDER_NAME (= "android.location.extr ...)	String
f	GPS_PROVIDER (= "gps")	String
f	KEY_LOCATION_CHANGED (= "location")	String
f	KEY_PROVIDER_ENABLED (= "providerEnabled")	String
f	KEY_PROXIMITY_ENTERING (= "entering")	String
f	MODE_CHANGED_ACTION (= "android.location.MODE ...)	String
f	NETWORK_PROVIDER (= "network")	String
f	PASSIVE_PROVIDER (= "passive")	String
f	PROVIDERS_CHANGED_ACTION (= "android.location.PROV ...)	String
f	KEY_STATUS_CHANGED (= "status")	String
	class	

Location Providers

Enable Location Services

```
final boolean gpsEnabled = locationManager
    .isProviderEnabled(LocationManager.GPS_PROVIDER) ;

private void enableLocationSettings() {
    Intent settingsIntent = new Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS) ;
    startActivity(settingsIntent) ;
}
```

LocationManager class

- Getting the last known position:
 - Location **getLastKnownLocation**(String source)
- Gets GPS receiver state;
 - GpsStatus **getGpsStatus**(GpsStatus status)
- Selecting the best provider based on a predefined criteria:
 - **getBestProvider**(Criteria criteria, boolean activeProviders)

LocationListener Interface

- Receiving updates when location changes
 - Based on frequency, distance
- Main approach is by using the override method:
 - void **onLocationChanged**(Location position)
 - Access coordinates by using the **Location type** object

Location Listener

```
LocationListener locationManager = new LocationListener() {  
    @Override  
    public void onLocationChanged(@NonNull Location location) {  
  
    }  
};
```

Request Updates

- **requestLocationUpdates()** is used for getting the current location
- Parameters:
 - The minimum time for querying updates
 - The minimum distance for querying updates
- Other parameters:
 - Location source:
 - **GPS_PROVIDER** or **NETWORK_PROVIDER** (LocationManager)
 - Selection criteria
 - The **LocationListener** type object
- **requestSingleUpdate()**

Location class

- Coordinates
 - **getLatitude(), getLongitude(), getAltitude()**
- Speed
 - **getSpeed()**
- Time
 - **getTime()**
- Location accuracy
 - **getAccuracy()**
- Other infos

Current position example

// create an object whose class implements LocationListener

```
LocListener locListener = new LocListener();
```

//getting the location service

```
LocationManager locManager = (LocationManager) getSystemService(Context.LOCATION_SERVICE);
```

//requesting the current location from a location provider

```
locManager.requestLocationUpdates(LocationManager.GPS_PROVIDER, 0, 0, locListener);
```

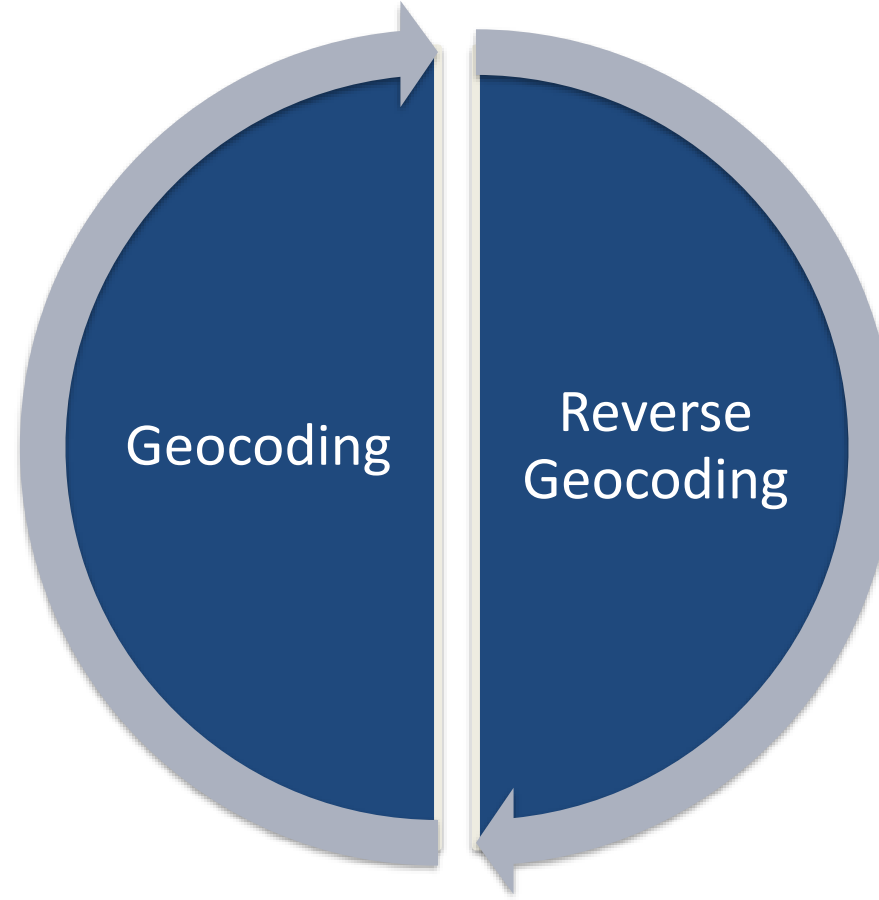
//removing the associated listener

```
locManager.removeUpdates(locListener);
```

Geocoder class

Google Inc.

1600 Amphitheatre
Parkway
Mountain View, CA 94043
Phone: +1 650-253-0000



Latitude:

37.422844

(37°25'21.3"N)

Longitude:

-122.084342

(122°05'03.8"W)

Geocoder class

- Backend services
- Must check the availability of the service
 - static method `isPresent()`
- Constructor initialization
 - receives the Context of the application as single parameter
- INTERNET permission required

Getting addresses

- Methods
 - `getFromLocation(lat, long, noMaxAddresses)`
 - `getFromLocationName()`
- Address Lists
 - Address class
 - `List<Address>`
- **Address** class
 - `getCountry()`
 - `getCountryCode()`
 - `getLocality()`
 - `getAddressLine()`

Getting addresses

```
if (Geocoder.isPresent()) {  
    Geocoder geocoder = new Geocoder(getApplicationContext());  
    try {  
        List<Address> addresses = geocoder.getFromLocation(latitude, longitude, 1);  
        if (addresses != null && addresses.size() > 0) {  
            for(Address address : addresses) {  
                StringBuilder builder = new StringBuilder();  
                for (int i = 0; i < address.getMaxAddressLineIndex(); i++)  
                    builder.append(address.getAddressLine(i));  
                builder.append("\n");  
            }  
        }  
    }  
    catch (IOException ex) {  
        ex.printStackTrace();  
    }  
}
```

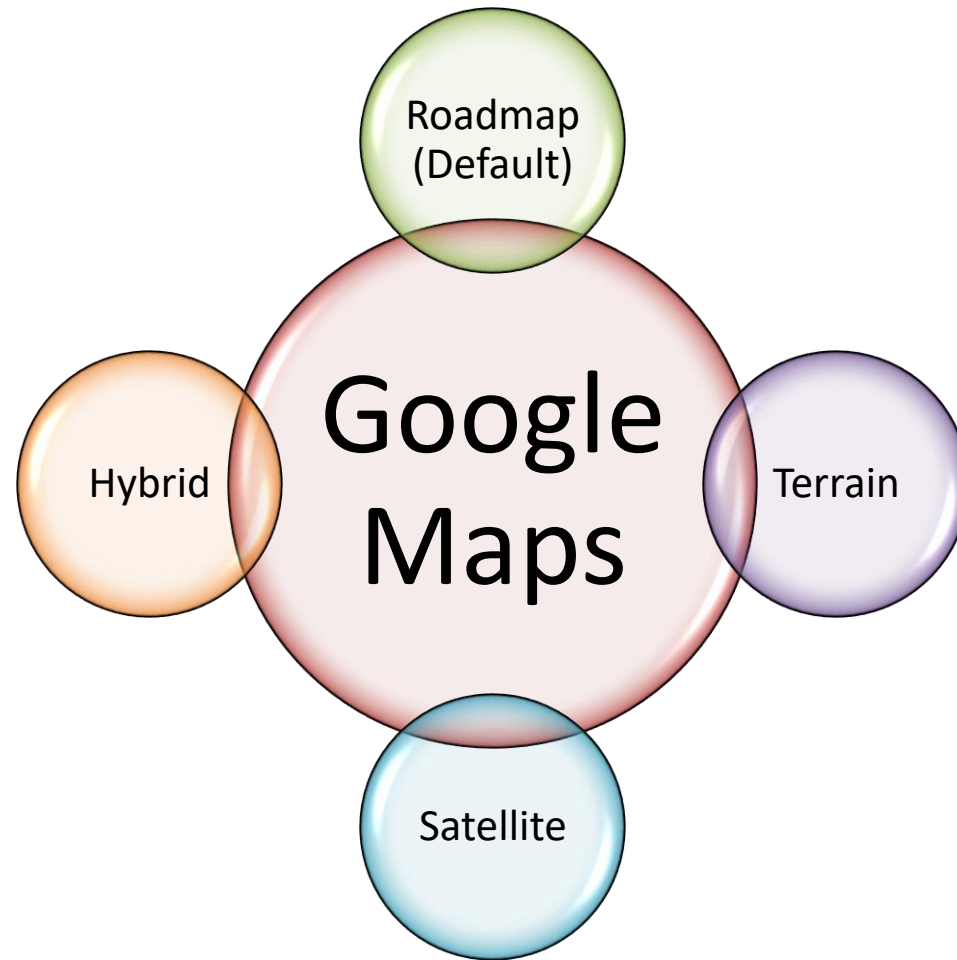
The image features the Google Maps logo, which consists of a stylized map of the world in shades of blue and green. The logo is centered within a large, irregular, dark blue ink splash that has a textured, painterly appearance. The splash is set against a white background with scattered blue ink droplets and splatters. The text "GOOGLE MAPS" is written in a bold, white, sans-serif font across the center of the splash.

GOOGLE MAPS

Google Maps

- Google Maps API
- Based on Google Maps service
- Included in *Google Play services*
 - Found in the Extras category from SDK Manager
- *Google Play services APK must be present on the mobile device*
- Instruments:
 - 3D maps, markers, paths, etc.

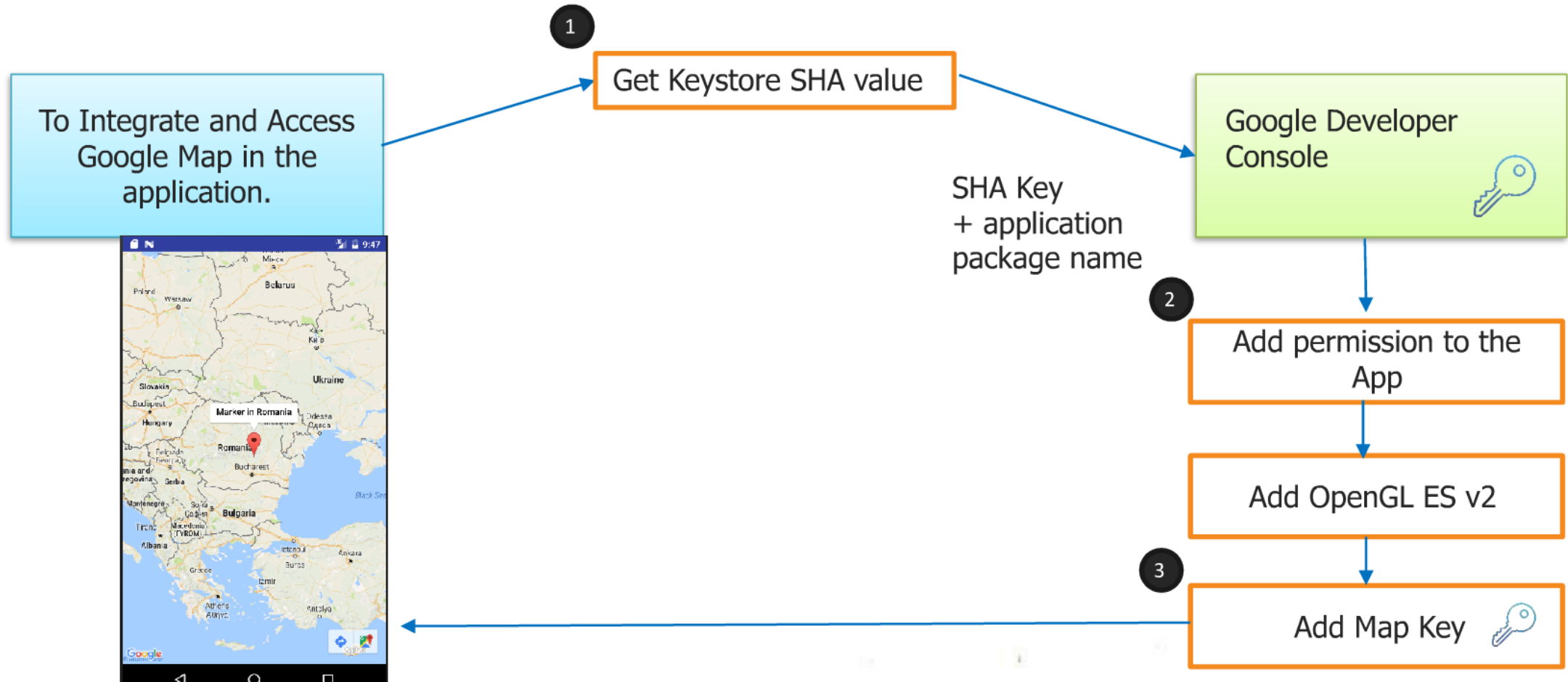
Google Maps



Initialization

- Getting a valid Maps API key from Google Maps Services
 - <https://code.google.com/apis/console>
 - The key must be inserted into a valid XML resource
- Installing the *Google Play services* SDK
- Adding the *Google Play services* libraries
- Using the *Google Play services* libraries

Google Maps



Getting the API Key

To get one, follow this link, follow the directions and press "Create" at the end:

`https://console.developers.google.com/flows/enableapi?apiid=maps_android_backend&keyType=CLIENT_SIDE_ANDROID&r=E5:96:2C:F8:01:5F:04:99:62:81:3B:82:31:FC:6C:7F:D1:B8:E4:6F%3Bcom.example.mihai.maps`

You can also add your credentials to an existing key, using this line:

`E5:96:2C:F8:01:5F:04:99:62:81:3B:82:31:FC:6C:7F:D1:B8:E4:6F;com.example.mihai.maps`

Alternatively, follow the directions here:

`https://developers.google.com/maps/documentation/android/start#get-key`

Once you have your key (it starts with "AIza"), replace the "google_maps_key" string in this file.

Google Maps API

- `com.google.android.gms.maps` package
- **MapView**
 - A control which encapsulates a map
- **MapFragment**
 - The fragment object used for displaying the map
- **UiSettings**
 - Control for setting the map preferences

Google Maps

- Map control
- Initialization
 - MapFragment#**getMap()** or MapView#**getMap()**
- Camera control
 - **moveCamera(CameraUpdate)**
 - **CameraUpdateFactory**
- Map type
 - **setMapType(mapType const)**
 - MAP_TYPE_NORMAL, MAP_TYPE_SATELLITE, MAP_TYPE_TERRAIN
- Settings
 - **getUiSettings()**

Google Maps

Circle

- Circle addCircle(CircleOptions)

Lines

- Polyline addPolyline(PolylineOptions)

Images

- GroundOverlay
addGroundOverlay(GroundOverlayOptions)

Polygons

- Polygon addPolygon(PolygonOptions)

Google Maps API v2 classes

- **Marker**

- Small items on the map showing certain locations
- Properties
 - Title
 - Position (coordinate) – given as **LatLng (latitude and longitude)**
 - Icon
 - etc.
- `GoogleMaps#addMarker(MarkerOptions)`

- **MarkerOptions**

- Setting the Marker options

Google Maps

- Map fragment declared as an xml layout file (res/layout)
 - fragment
 - android:id="@+id/map"
 - android:name="com.google.android.gms.maps.MapFragment"
- The java class associated file:

```
setContentView(R.layout.activity_maps);  
MapFragment mapFragment = (MapFragment)  
    getSupportFragmentManager().findFragmentById(R.id.map);  
mapFragment.getMapAsync(this);
```

Google Maps

- AndroidManifest.xml

- Minimum set of permissions:

- `<uses-permission
 android:name="android.permission.ACCESS_FINE_LOCATION" />`

- Google Maps API key

```
<meta-data  
    android:name="com.google.android.geo.API_KEY"  
    android:value="@string/google_maps_key" />  
<string name="google_maps_key">  
    AIzaSyeqGdwidSbgQFTjdTpzZBZ6kN7gB8I_Jd0  
</string>
```

Google Maps Markers

Following is the code snippet for showing Simple Marker on Google Map

```
mGoogleMap.addMarker(new MarkerOptions()  
    .title("Awesome title")  
    .snippet("The most populous city")  
    .position(latLng));
```

Customizing the Marker opacity

```
mGoogleMap.addMarker(new MarkerOptions()  
    .title("Awesome title")  
    .snippet("The most populous city")  
    .position(latLng)  
    .setAlpha(0.7f);
```

Google Maps Markers

Customizing Marker Image

```
mGoogleMap.addMarker(new MarkerOptions()  
    .title("Awesome title")  
    .snippet("The most populous city")  
    .position(latLng)  
    .alpha(0.7f)  
    .icon(BitmapDescriptorFactory.fromResource(R.drawable.abc_btn_check_material)));
```

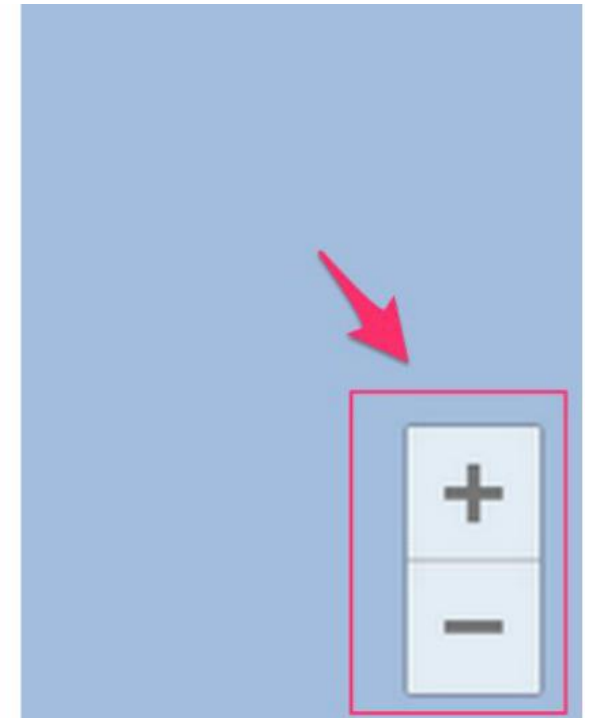
Rotate the Marker

```
mGoogleMap.addMarker(new MarkerOptions()  
    .title("Awesome title")  
    .snippet("The most populous city")  
    .position(latLng)  
    .alpha(0.7f)  
    .icon(BitmapDescriptorFactory.fromResource(R.drawable.abc_btn_check_material))  
    .rotation(90.0f));
```

Zoom in/out

Zoom Control

```
@Override  
public void onMapReady(GoogleMap googleMap) {  
    mGoogleMap = googleMap;  
    mGoogleMap.getUiSettings().setZoomControlsEnabled(true);  
    showMyLocationOnScreen();  
}
```

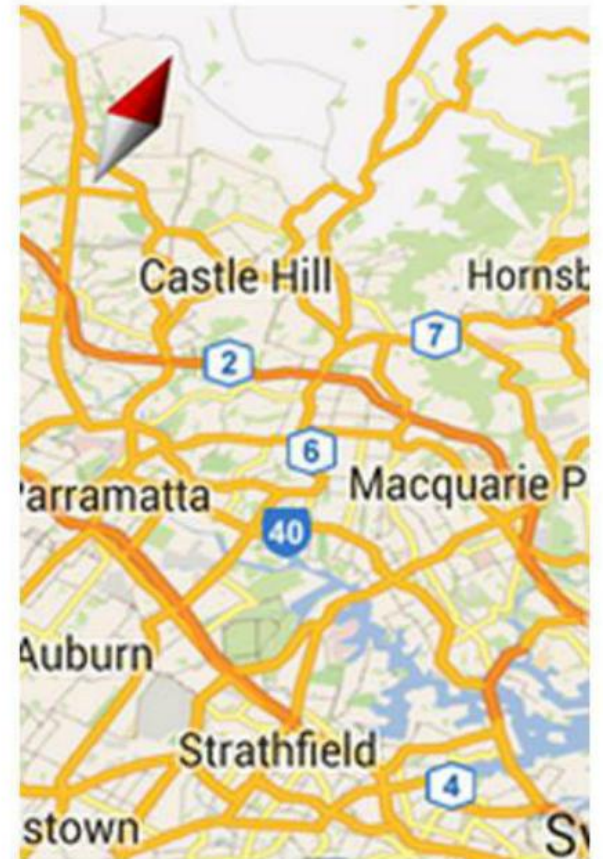


Compass Control

Compass Control

```
@Override
public void onMapReady(GoogleMap googleMap) {
    mGoogleMap = googleMap;
    mGoogleMap.getUiSettings().setZoomControlsEnabled(true);
    mGoogleMap.getUiSettings().setCompassEnabled(true);

    showMyLocationOnScreen();
}
```



My Location Button

My Location Button

```
@Override
public void onMapReady(GoogleMap googleMap) {
    mGoogleMap = googleMap;
    mGoogleMap.getUiSettings().setZoomControlsEnabled(true);
    mGoogleMap.getUiSettings().setCompassEnabled(true);
    mGoogleMap.getUiSettings().setMyLocationButtonEnabled(true);
    mGoogleMap.setMyLocationEnabled(true);
    mGoogleMap.setOnMyLocationChangeListener(this);

    showMyLocationOnScreen();
}
```



Resourcess

- S. Komatineni, D. MacLean – Pro Android 4, Apress, 2012
- R. Meier – Professional Android 4 Application Development, Wiley, 2012
- P. Pocatilu, Programarea dispozitivelor mobile, Editura ASE, 2012
- P. Pocatilu, I. Ivan ș.a. – Programarea aplicațiilor Android, Editura, ASE, 2015
- <http://developer.android.com>